

Unitree G1 EDU Ultimate — Pelvis Thermal / Odor Fault

Prepared for Unitree Technical Support | Date: 2026-07-14 | Unit: G1 EDU Ultimate (29 DoF + Inspire FTP hands) | Serial: _____

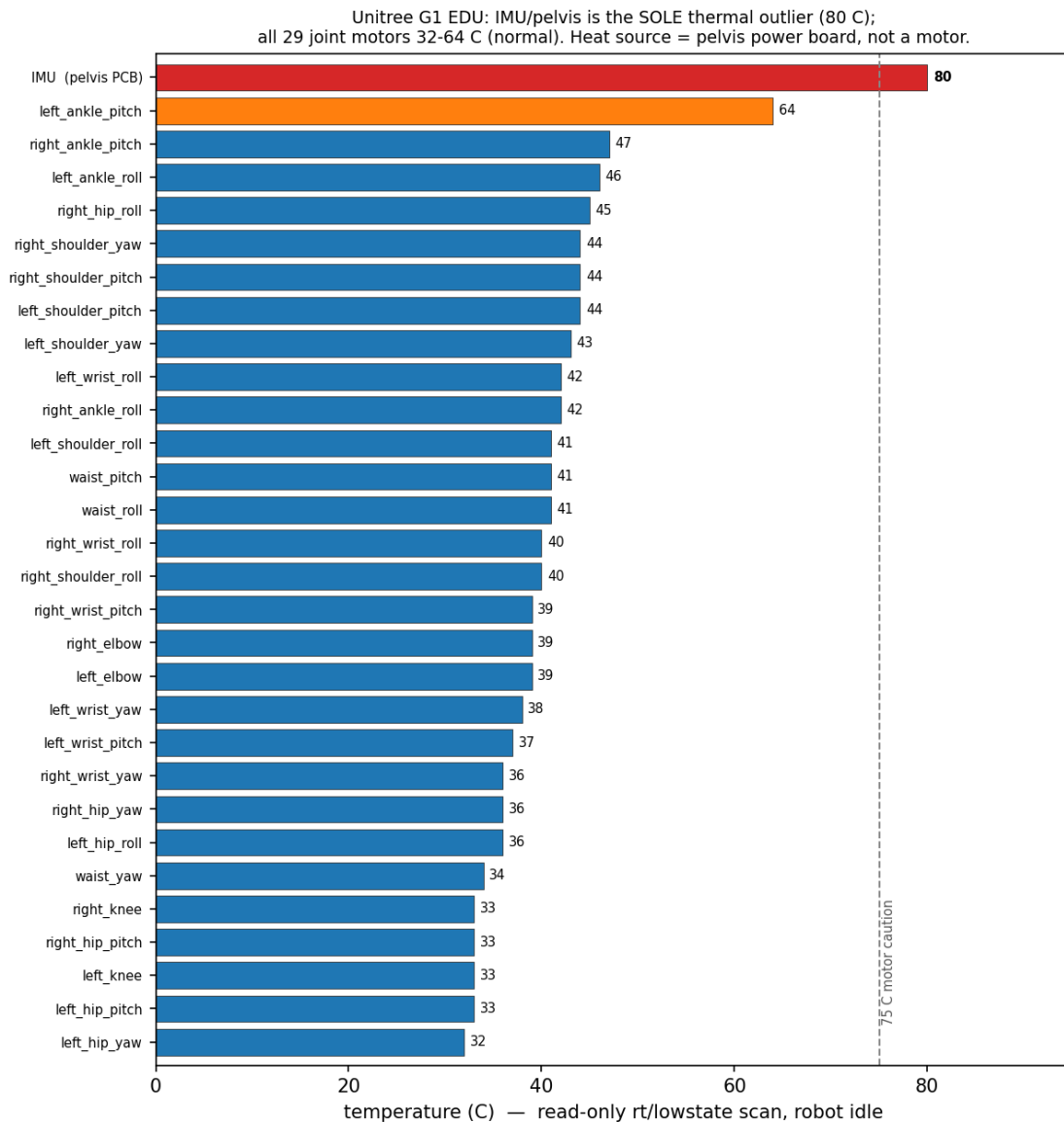
1. Symptom

A persistent burning / rotten (fishy) odor is localized to the **pelvis / waist enclosure**. The odor has continued for ~24 hours and is present **even with the robot powered off**. The robot otherwise operates normally (stands, holds pose, all joints respond). During operation the **on-board IMU reports 80 °C**, while every joint actuator remains at a normal temperature.

2. Measured evidence (read-only telemetry)

Captured over DDS rt/lowstate (subscribe-only; no commands sent), robot idle:

- **IMU temperature: 80 °C** — the single thermal outlier.
- **All 29 joint motors: 32–64 °C (normal)**. Hottest = left_ankle_pitch at 64 °C (expected balance load). Pelvis-area motors are cool: waist yaw/roll/pitch = 34/41/41 °C; hips 33–45 °C.
- Across **54 recorded runs over 6 days**, motor temperatures were consistently normal (34–64 °C); no motor ever exceeded 64 °C. The IMU / pelvis region is the only hot area.



3. Interpretation

The IMU sensor dissipates only milliwatts and cannot self-heat to 80 °C; the reading therefore reflects the **temperature of the pelvis PCB region around the IMU**. With all motor windings normal, the heat source is the **pelvis**

power-distribution electronics — not a joint/actuator or a mechanical bind. The odor persisting after power-off indicates a component has **already partially degraded** (e.g. vented capacitor residue), not merely an over-temperature transient.

4. Most likely causes (ranked, by our assessment)

#	Suspected cause	Reasoning
1	Vented / degrading electrolytic capacitor on the pelvis / power board	Pelvis / power board odor persists after power-off (electrolyte residue); localized heat.
2	Failing DC-DC converter stage (buck regulator ET in simulator)	ET in simulator an always-on rail (explains heat even at idle); hot-varnish odor.
3	Overheating / out-of-spec resistor charring the PCB	PCB odor + localized hotspot.
4	High-resistance power connector (loose/corroded Contact)	Contact heating at a connector; hot-plastic odor.
5	BMS / battery-path component	Lower likelihood, flagged for safety (Li-ion path).

5. Ruled out / actions taken

- Motor windings ruled out (normal temperatures across all 54 runs).
- Not operated under high load (no dance / whole-body control) since onset; battery removed between sessions.
- No liquid ingress or impact event preceded the fault.

6. Requests to Unitree

- Is this a **field-serviceable** pelvis power-board component, or does it require **RMA / service return**?
- If this is a known issue on this hardware revision, please provide the specific part and procedure.
- Please confirm whether the unit is **safe to operate under load** until resolved (our assessment: it is not).
- If RMA is required, the process for an EDU Ultimate unit.

Appendix A — full joint temperature scan (max °C, idle)

IMU (pelvis PCB): 80	left_hip_pitch: 33	left_hip_roll: 36
left_hip_yaw: 32	left_knee: 33	left_ankle_pitch: 64
left_ankle_roll: 46	right_hip_pitch: 33	right_hip_roll: 45
right_hip_yaw: 36	right_knee: 33	right_ankle_pitch: 47
right_ankle_roll: 42	waist_yaw: 34	waist_roll: 41
waist_pitch: 41	left_shoulder_pitch: 44	left_shoulder_roll: 41
left_shoulder_yaw: 43	left_elbow: 39	left_wrist_roll: 42
left_wrist_pitch: 37	left_wrist_yaw: 38	right_shoulder_pitch: 44
right_shoulder_roll: 40	right_shoulder_yaw: 44	right_elbow: 39
right_wrist_roll: 40	right_wrist_pitch: 39	right_wrist_yaw: 36

Data source: read-only DDS rt/lowstate thermal scan, 2026-07-13. Raw log available on request.